



City-scale assessment of outdoor heat shelter availability under climate change conditions: a simulation-based methodology

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ABSTRACT

The paper proposes a new methodology for assessing the availability of outdoor heat shelters in cities during future hot summer days. Heat shelters are identified on 1 meter resolution physiological equivalent temperature (PET) fields obtained with SOLWEIG simulations. Input weather data are representative hot summer days (RHSDs) which are processed from an analysis of future weather projections, for three stages of climate change evolution. The heat shelter availability is assessed through three indicators: the heat sheltering status, the heat shelter area per capita, and the heat sheltering performance. In this way, the methodology makes it possible to assess the time evolution of the heat shelters availability, over the course of the day and over the century, in large urban areas, for different levels of protection against heat. The methodology is applied to the city of Lyon. In this city, heat shelters will not provide thermal comfort; they will at most protect against heat stress exposure. During the core of the day, at the end of the century, heat shelters are exclusively located in shaded areas with sufficient ventilation and low exposure to reflected solar radiation. Locations under large patches of continuous tree cover or at the foot of building façades without overlooks fulfill these conditions. In the parks, the daytime average heat shelter area is proportional to the surface of the tree canopy.

1. Introduction

During hot summer days, urban residents may be exposed to excessively hot outdoor environments. These hot environments may cause discomfort or health issues. Exposure to excessive outdoor heat is expected to occur more often and with a higher intensity in the future than today [28].

Extreme heat exposures occur during the day. They are due to the combination of high air temperature, and strong solar radiation. Because of the solar radiation contribution, heat exposure conditions in the outdoor environment are highly heterogeneous, with significant variations over short distances. A pedestrian can experience very different thermal exposures between two locations separated by just one meter, if one of these locations is exposed to direct sunlight and the other is shaded.

The present paper focuses on cities for which the urban heat island (UHI) effect may be neglected during daytime. The UHI is a physical phenomenon that causes differences in air temperature between urban and adjacent non-urban sites. This difference in air temperature is referred to as the urban heat island intensity (UHII). The UHI is known to

be a nighttime phenomenon [58]. A typical daily UHII pattern was established by [62] based on measurements in Vancouver: the UHII reaches a few degrees at night, it is almost zero and sometimes negative for most of the day. It decreases rapidly after sunrise and increases slowly in the late afternoon. This pattern has been observed in several cities around the world: Berlin [12], Rennes (France) [13], Singapore [8], etc. For these cities, extreme heat exposures are not due to the UHI, as UHII is almost zero when extreme heat exposures occur, i.e., during daytime. To the authors' knowledge, significant daytime UHII values have been reported only in coastal cities affected by sea breezes: Athens [14], Shanghai [21], etc. For these cities, the UHI must be taken into account when studying extreme heat exposures. These cities are out of scope for the present study.

Heat shelters [23,33], also referred to as "climate shelters" [1,44] or "cooling shelters" [37], are locations that are accessible to the population, and in which the thermal environment remains bearable during hot summer days. Heat shelters can be part of a city's adaptation strategies, as in Barcelona [59]. Heat shelters may be located within buildings (churches, shopping malls, etc.), or in the outdoor environment (green infrastructures, etc.). The present paper only focuses on heat shelters

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located in the outdoor environment.

Indoor and outdoor heat shelters have been investigated in studies based on vulnerability [1,9,33,44,59,67]. But outdoor heat shelters have never been studied from the point of view of thermal performance: their identification has never been based on a prediction of thermal indices that would enable us to assess their ability to provide acceptable thermal conditions. The present study intends to fill this gap, by proposing a methodology for identifying heat shelters based on simulations of outdoor thermal environments.

The evaluation protocol is designed to provide answers to the following questions:

1. Are there enough heat shelters to meet the needs of the population?
2. What level of protection can be expected in heat shelters?
3. How does the availability of heat shelters evolve during the daytime and during the century?
4. What are the specificities of the urban fabric at the location of the heat shelters?
5. Are the heat shelters equitably distributed over the city?

To answer question 1, the evaluation must be made over a sufficiently large area so that the availability of the shelters could be related to the number of occupants. Indeed, if the study domain is too small, there is a high probability that inhabitants may move across the boundaries of the domain and find heat refuges elsewhere.

In question 2, two levels of protection must be considered: protection against thermal discomfort and protection against exposures to heat stress that may cause health issues. For question 3, two time scales are distinguished: the evolution of heat shelter availability during the daytime and the evolution of heat shelter availability over the century. These changes are both due to the evolution of meteorological patterns. Answering questions 2 and 3 requires simulating the thermal behavior of the urban fabric under various meteorological conditions. Answering question 4 requires a qualitative analysis of the heat shelter distribution over the studied domain. Answering question 5 requires independently assessing the availability of heat shelters over several portions of the city.

Answering questions 1, 2 and 3 requires simulating the thermal behavior of the urban fabric under various meteorological conditions, and over large urban domains. The simulation tools that are devoted to outdoor thermal exposure studies are the micrometeorological simulation models. To capture the sharp contrasts in the outdoor thermal environment, the micrometeorological models predict the values of thermal environment variables (air temperature, solar and infrared radiation, wind speed and relative humidity) at a high spatial resolution, usually less than 1 meter. Envi-Met [30,35,57] and Solene Micro-Climat [19,52,53] are two of the most commonly encountered micrometeorological simulation models. But these models combine a CDF model and a detailed radiation model that require a lot of computer resource. They are not suitable for simulations that cover a large domain.

The present paper proposes a heat shelter evaluation protocol that permits to provide answers for the five questions listed above. The protocol can be applied over urban zones that span over several dozen square kilometers. It makes use of the micrometeorological model Solar and LongWave Environmental Irradiance Geometry (SOLWEIG), which has low computational demands [15,39,41], to predict the behavior of the urban layout under various meteorological conditions.

The evaluation protocol will be presented in the first part of the paper. Then, it will be applied to a case study, for which the location of interest is a portion of the Lyon conurbation (France). The description of the area of interest and the objectives of the case study will be presented in the second section of the paper. The results will be presented and discussed in the following sections.

2. Evaluation protocol

2.1. Overview of the evaluation protocol

Fig. 1 shows a schematic description of the heat shelter evaluation protocol. The weather files for three representative hot summer days (RHSDs) are first produced. Then, the distributions of the mean radiant temperature t_{rm} over the studied domain during the three RHSDs are computed with SOLWEIG. The heat shelters cannot be identified based on the mean radiant temperatures because there is no clear mean radiant temperature threshold to discriminate between acceptable and non-acceptable thermal environments. This is why distributions of Physiological Equivalent Temperature (PET) are computed in the next step. Then, the heat shelters are identified. Three indicators are produced and analyzed to characterize and quantify the availability of heat shelters over the studied domain: the heat shelter status $\hat{I}(\vec{x}, t)$ over the domain, the heat shelter area per capita $\hat{A}(t)$, and the heat sheltering performance of the urban fabric $\hat{\eta}$.

All the data manipulations are performed with the software QGIS. The raster manipulation tools mentioned in the figure are part of the QGIS processing toolbox, they include tools from the GDAL library such as the raster calculator. The next sections provide more detail about each step of the protocol.

2.2. Representative hot summer day weather data

2.2.1. State of the art

The production of representative future weather data is an active research subject in various fields, such as building or urban layout performance assessment [73]. Several techniques that have been developed for producing representative future weather data make use of weather projections. A weather projection is a time series of meteorological data that covers the whole 21st century. A weather projection is produced by a global circulation model (GCM) and a regional climate model (RCM) for one representative concentration pathway (RCP) scenario. The discrepancies between the weather projections are due to the input RCP scenarios and to the models [20]. Weather projections may be provided by national research institutions or downloaded from the database of the CORDEX project [17,29]. Representative future weather data are either directly extracted from weather projections or artificially generated using a morphing method [60].

Representative future weather data may cover full years [10,60,70,71], the duration of extreme weather events [64,72], or single days. Simulating a single day is sufficient to assess outdoor heat exposures during the day [30,32,43,45,55,61,68]. Thus, for daytime outdoor thermal environment assessment, representative future weather data may be restricted to single representative future days.

The local weather types identification is an advanced technique for producing single day representative weather data [22]. The method first identifies representative weather conditions by performing a cluster analysis on daily mean weather variables. Then it reconstructs time series of weather data from the medoid of each cluster. In its current form, this method is not suitable to most studies about daytime heat exposure. These studies intend to assess the thermal exposure during particularly hot days. The weather data should be differentiated, at least during one stage of the selection process, through a unique indicator that measures the level of heat exposure expected from the weather conditions. This is a condition for controlling the relative warmth of the selected days.

The literature only reports two examples of selections of representative future days specifically dedicated to daytime outdoor heat exposure studies. Karner et al. [32] selected one particularly hot day within a weather database for the San Francisco Bay Area. Their selection method was based on a quantile of the daily average temperature. However, the representative future day was selected among historical

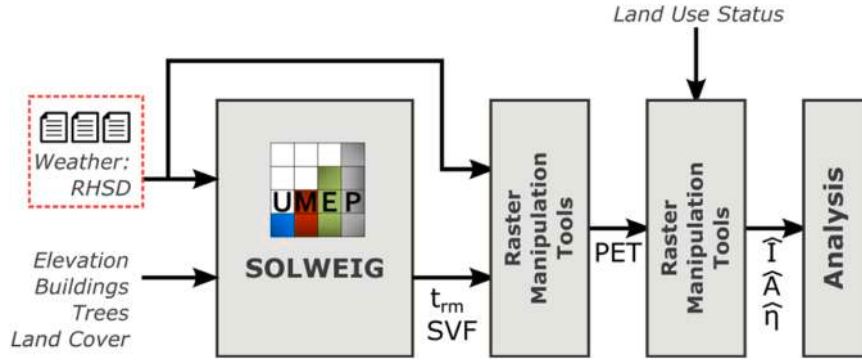


Fig. 1. Schematic description of the heat shelter evaluation protocol.

weather data instead of projected weather data. Mahmoud & Ragab [45] picked weather data for July 13, 2035, in the city of Aswan (Egypt) from two weather projections: one for the RCP4.5 scenario and one for the RCP8.5 scenario. Both weather projections were produced by the same model. The choices of the date (July 13) and the year (2035) were not justified.

Therefore, no fully rational weather processing technique has ever been applied to produce representative future days from future weather projection data, in the context of daytime heat exposure studies. The next section proposes such a rational weather processing technique.

2.2.2. Input weather processing technique

The objective of the weather processing technique is to produce, from a weather projection database, three representative hot summer days (RHSDs) for three stages of evolution in climate change over the century: the current state (2010–2040), the mid-century stage (2040–2060), and the end-century stage (2080–2100). The representative hot summer days must correspond to particularly hot days, but not to extreme days in terms of overheating. No distinction is made between the RCP scenarios, these scenarios are assumed to be equiprobable.

To represent the whole diversity of possible future weather conditions in the city of interest, the input database must gather weather projections from various GCM+RCM models and for various RCP scenarios. It must be balanced: no model or RCP scenario should be overrepresented.

For each stage of the evolution of climate change:

- A subset of daily weather data is constituted with all days falling in July during the period of interest.
- The distribution of the daily maximum air temperature $t_{a,dmax}$ is calculated, and the days for which $t_{a,dmax}$ lies between the 75th and the 85th percentile of the distribution are selected.
- Within the selected days, the median value of the following daily statistics is calculated: daily minimum and maximum air temperature, daytime average value of wind speed, specific humidity, difference between air and sky temperature, and global horizontal solar radiation. The daytime period over which averages are calculated is the period during which the sun is above the horizon.
- The daily statistics are used to reconstruct a time series of meteorological data over a whole day: this is the representative hot summer day.

The following procedure is employed to reconstruct the time series of weather data. Global, diffuse and direct clear sky solar radiation evolutions, for the location of interest and for July 14th, are first produced using the pvLib Python library [2,26]. The solar radiation values are then scaled to adjust the daytime average global horizontal solar radiation to the selected median value. The air temperature evolution is constructed with the selected daily minimum and maximum air temperatures, using the technique described in [50]. The daily patterns of

the remaining weather variables are constructed by considering that the wind speed, the difference between air and sky temperature, and the specific humidity are constant and equal to the selected median values.

2.2.3. Comments

With the technique described above, the daily maximum air temperature of the RHSDs is the 80th percentile of the daily maximum air temperatures of the days in July during the period of interest. The choice of July is the result of a compromise between high air temperature and high solar radiation values. In the Northern Hemisphere, the solar radiation peaks on June 21. In France, the highest air temperatures are measured at the end of July [49].

The choice of the 80th percentile may be interpreted as follows: there is 1 chance out of 5 that the outdoor weather conditions would be hotter than the RHSD on a July day. The RHSD thus does not represent extreme conditions but conditions that may occur repeatedly.

The other daily statistics are selected independently, after filtering the data between the 75th and the 85th percentile of $t_{a,dmax}$ among the days in July. The purpose of this process is to select a combination of weather data that is realistic, without being too specific to a particular day.

2.3. Mean radiant temperature computation

The distribution of the mean radiant temperature over the studied domain is computed with the SOLWEIG model. The basic principles of the SOLWEIG model are explained by Lindberg et al. [41]. Since 2018, the model has gone through two major updates, with the inclusion of a submodel for the trees [39] and the capacity to handle various ground albedos [42]. SOLWEIG has gone through several validation studies [7, 15,36,41]. It has been implemented as a part of the Urban Multiscale Environmental Predictor (UMEP) suite for the QGIS software [40].

The mean radiant temperature is defined with the following equation:

$$t_{rm} = \left(\frac{\epsilon_{cl} \cdot L^l + \alpha_{cl} \cdot S^l}{\sigma \cdot \epsilon_{cl}} \right)^{\frac{1}{4}} \quad (1)$$

In this equation, L^l and S^l are, respectively the longwave and the shortwave radiation flux densities received by the pedestrian from all the directions of space (in W/m^2). $\sigma = 5.67 \cdot 10^{-8} Wm^{-2}K^{-4}$ is the Boltzmann constant. ϵ_{cl} and α_{cl} are the emissivity and the solar absorption coefficient of the pedestrian's clothing, respectively.

To capture the sharp variations in t_{rm} within the studied domain, the simulations must be performed with a spatial resolution below or equal to 1m. To perform simulations over a large domain with a desktop computer, the domain can be split into smaller subdomains, and the simulation can be performed independently for each subdomain.

2.4. Remaining meteorological variables

The calculation of the *PET* requires three additional meteorological variables: air temperature t_a , relative humidity RH , and wind speed at pedestrian level v_a .

The air temperature is considered to be homogeneous all over the studied domain. As mentioned in the introduction, this study focuses on cities where UHI is a nocturnal phenomenon. The period of interest being the daytime period, spatial variations in air temperature due to UHI can be neglected during the period of interest. The protocol does not consider neither the variations in air temperature with the elevation due to adiabatic expansion of air. The specific humidity is considered as spatially homogeneous over the city. Since the air temperature and the specific humidity are homogeneous, the relative humidity is also homogeneous within the studied domain.

The spatial variations of wind velocity are estimated using a regression model. The regression model was obtained by fitting a third-degree polynomial equation (Eq. 2) to the reference values of wind velocity ratio (*WVR*) and sky view factor (*SVF*) generated by 3D wind flow simulations by Feng [11]. These data are drawn in Fig. 2. *SVF* ranges from 0 (fully enclosed areas) to 1 (open fields). The polynomial equation has been constrained to avoid discontinuities at both ends ($WVR'(0) = WVR'(1) = 0$), and to ensure that $v_a = 0\text{m/s}$ in fully enclosed areas ($WVR(0) = 0$). The fitting process led to $b = 2.22$. The resulting curve is drawn in Fig. 2. With the regression equation, the wind velocity ratio for open area is $WVR(1) = 0.74$. This value is very close to the ratio $\log(1.5/0.01)/\log(10/0.01)$, which expresses the velocity ratio between measurement height and screen height in open fields [6].

$$WVR = \frac{v_a}{v_{a,met}} = b \cdot SVF^2 - \frac{2}{3} b \cdot SVF^3 \quad (2)$$

The wind speed values predicted with the regression Eq. (2) will be bounded with a lower value of 0.15 m/s. This bounding value corresponds to wind speed that may be encountered in closed buildings due to natural convection or forced ventilation [46].

2.5. Physiological equivalent temperature (*PET*)

2.5.1. Definition and computation of the *PET*

The *PET* is defined as the air temperature at which, in a reference environment ($T_{mrt} = T_{air}$, $v_a = 0.1\text{m/s}$, $p_v = 12\text{ hPa}$), the heat balance of the human body (metabolic rate $met = 80\text{W}$, i.e., light activity; heat resistance of clothing $clo = 0.9$) is maintained with skin and core temperatures equal to those under the environmental conditions being

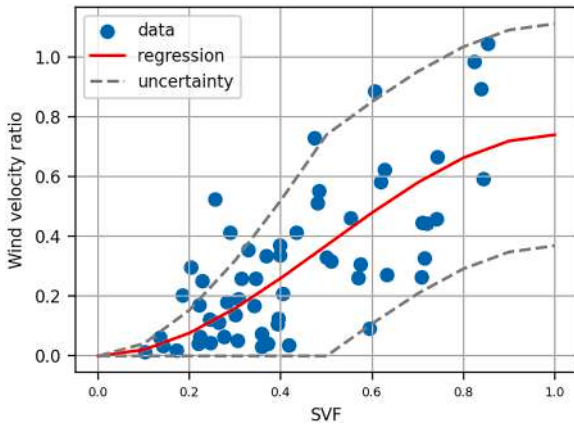


Fig. 2. Wind velocity ratio as a function of the sky view factor. Simulation results from [11] and the regression curve.

assessed [25]. It is probably the most widely used thermal index in outdoor heat exposure studies [43,45,51,55].

The *PET* is based on the Munich Energy-balance Model for Individuals (MEMI). When computing the *PET*, an iterative numerical scheme is needed to solve the physiological heat balance equation. The time for computing the *PET* over a large domain with high resolution is considerable. To speed-up the computation process, a regression approach is proposed. Since, for each RHSD, the specific humidity is considered as constant and homogeneous, the *PET* only depends on three independent meteorological variables that may vary in time and space: t_a , v_a and t_{rm} . After computing the mean radiant temperature fields with SOLWEIG, a dataset of *PET* values that covers the whole ranges of observed values for t_a , $t_{rm} - t_a$, and v_a is constituted. A multivariate polynomial regression is applied to the dataset. The resulting polynomial model allows for a quick computation of the *PET*. This process has to be repeated for each RHSD. It is similar to the approach employed for accelerating the computation of the Universal Thermal Climate Index (UTCI) [6].

2.5.2. Definition of the heat shelters based on *PET* thresholds

Table 1 presents the original assessment scale of the *PET* [48]. The identification of the heat shelters will be based on two thresholds given by this table.

The discomfort threshold for the present study is fixed at $PET^{dis} = 29^\circ\text{C}$. It is the limit between a slightly warm thermal sensation, and a warm thermal sensation. A field study carried out in two Brazilian and one English city showed that this comfort threshold leads to 50 % thermal satisfaction [34]. Another field study conducted in Hungary showed that 29°C was the *PET* value above which pedestrians began to prefer lower air temperature and solar radiation [31]. A location will be considered as a heat shelter for discomfort if $PET < 29^\circ\text{C}$.

The heat stress threshold is fixed at $PET^{hs} = 35^\circ\text{C}$. It is the transition between moderate and strong heat stress. This threshold has been quoted as the limit from which bioclimatic conditions affect health and well-being of people, especially in Europe [47]. A net increase in terms of mortality was observed in Vienna when $PET > 35^\circ\text{C}$ [54]. A location will be considered as a shelter for heat stress if $PET < PET^{hs}$.

2.6. Heat shelter-related indicators

2.6.1. Definitions

Three indicators are now computed from the distributions of *PET* (Fig. 3): the heat shelter status $\hat{I}(\vec{x}, t)$, the heat shelter area per capita $\hat{A}(t)$, and the heat sheltering performance of the urban fabric $\hat{\eta}$.

The heat shelter status is equal to 1 when the location of interest is a heat shelter and 0 otherwise. It is obtained by comparing the *PET* value to the threshold:

$$\hat{I}(\vec{x}, t) = \begin{cases} 1 & \text{if } PET(\vec{x}, t) \leq PET^{dis/oh} \\ 0 & \text{if } PET(\vec{x}, t) > PET^{dis/oh} \end{cases} \quad (3)$$

The heat shelter area per capita is the ratio between the heat shelter area $\hat{A}(t)$ and the number of inhabitants n_{hab} in the area of interest. The

Table 1
Original assessment scale of the *PET*.

<i>PET</i> [$^\circ\text{C}$]	Thermal perception	Thermal stress
< 4	Very cold	Extreme cold stress
4 to 8	Cold	Strong cold stress
8 to 13	Cold	Moderate cold stress
13 to 18	Cool	Slight cold stress
18 to 23	Neutral	No thermal stress
23 to 29	Slightly warm	Slight heat stress
29 to 35	Warm	Moderate heat stress
35 to 41	Hot	Strong heat stress
> 41	Very hot	Extreme heat stress

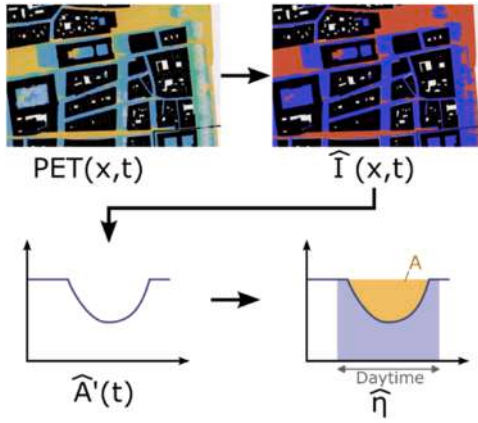


Fig. 3. Indicators related to the heat shelters.

heat shelter area is the spatial integration of the heat shelter status over the domain of interest:

$$\hat{A}(t) = \int \int \hat{I}(\vec{x}, t) \cdot dS(\vec{x}) \quad (4)$$

$$\hat{A}'(t) = \frac{\hat{A}(t)}{n_{hab}} \quad (5)$$

The daily evolution of the heat shelter area per capita $\hat{A}'(t)$ should look similar to the curve in Fig. 3. During the night, the PET is expected to remain below the threshold in the whole domain. $\hat{A}'(t)$ is constant and equal to the surface of the domain divided by the population A/n_{hab} . This is the “fully sheltered” state. During the daytime, solar radiation and air temperature reduce the availability of heat shelters: $\hat{A}'(t)$ decreases and reaches a minimum value during the afternoon.

The heat sheltering performance $\hat{\eta}$ is a number between 0 and 1. It measures the capacity of the urban fabric to provide heat shelters during the daytime, despite the intensity of the meteorological heat loads. It is defined as the daytime average value of the heat shelter area $\hat{A}(t)$ divided by the area of the domain A :

$$\hat{\eta} = \frac{\int_{daytime} \hat{A}(t) \cdot dt}{A \cdot \Delta t_{daytime}} \quad (6)$$

The daytime average heat shelter area per capita is the average value of the heat shelter area per capita $\hat{A}'(t)$ during the daytime:

$$\hat{A}'_{DT} = \frac{\int_{daytime} \hat{A}'(t) \cdot dt}{\Delta t_{daytime}} \quad (7)$$

2.6.2. Interpretation of the indicator values

The three indicators were designed so that an analysis of their values could provide answers to the five questions listed in the introduction.

The heat shelter area per capita $\hat{A}'(t)$ should indicate whether there are enough heat shelters for the population (Question 1) at a given time. The evolution of $\hat{A}'(t)$ during the three RHSDs should indicate how the availability of heat shelters evolves during the day and during the century (Question 3). The heat shelter area per capita can be computed for two levels of protection: protection against heat discomfort $\hat{A}'_{29}(t)$ and protection against heat stress $\hat{A}'_{35}(t)$. If the heat shelter area per capita is null during the daytime, the associated level of protection cannot be expected during hot summer days (Question 2). The maps of heat shelter status $\hat{I}$ will be useful for locating the heat shelters over the studied domain and determining the nature of the urban fabric around the heat shelters (Question 4).

The heat shelter area per capita $\hat{A}'(t)$ and the heat sheltering performance $\hat{\eta}$ can be computed over the whole studied domain or over several portions of the domain. Computing $\hat{A}'(t)$, $\hat{A}'_{DT}$ and $\hat{\eta}$ for different subdomains will allow us to evaluate whether the heat shelter distribution over the domain is equitable (Question 5). The heat sheltering performance indicator $\hat{\eta}$ and the daytime average heat shelter area $\hat{A}'_{DT}$ facilitate the comparison of the heat shelter availability between the various subdomains and RHSDs.

In the present paper, we propose to analyze the values of the indicators in three steps. The first step is a global analysis, which consists of analyzing the evolution of $\hat{A}'(t)$ and $\hat{\eta}$, computed for the whole domain, to obtain answers for questions 1, 2, and 3 at the global scale. The second step, which intends to answer question 4, is a qualitative analysis of the heat refuge maps over the studied domain. It is not necessary to analyze all the simulation results (the whole domain, during the three RHSDs, for the two levels of heat protection). The qualitative analysis can be restricted to configurations that highlight the discrepancies of the thermal response of various urban layouts. The last step, which intends to answer question 5, is a comparison of the heat shelter availability over various subentities of the studied domain. This three-step analysis will be applied to the case study presented in the next section.

3. Description of the case study

3.1. Description of the domain

The case study takes place in the conurbation of Lyon, France. The boundaries of the studied domain are drawn in Fig. 4. The studied domain is a central portion of the conurbation of Lyon. It covers 26 km² and hosts approximately 225,000 inhabitants. The eastern boundary of the domain is the Rhône River. The Saône River flows from north to south around the central meridian of the domain. The studied domain contains 8 municipalities: five districts of Lyon (Lyon 1, 2, 4, 5) and three peripheral cities (Caluire, Sainte-Foy, La Mulatière). The western parts of Sainte-Foy and Lyon 5 were removed from the domain. The heat shelter evaluation will only focus on outdoor public spaces.

The sources for the data drawn in Fig. 5 are listed in Table 1. Fig. 5 (a) shows the footprint and the height of the buildings, and the area covered by tree crowns. The description of the street trees is available in the Data Grand Lyon platform. The remaining trees were added manually, using Google Maps aerial photography [18] as reference. Isolated trees were added individually. Continuous tree canopies were reproduced with a random point generator. The trees that are located in private spaces (grey patches) were not collected. The core of the city is characterized by a dense population of midrise (approximately 24 m tall) buildings. The green patches on both sides of the Saône River correspond to the Balmes Lyonnaises (identified with letter B). The Balmes Lyonnaises consist mostly of green wastelands, which are closed to the public for safety reasons.

Fig. 5 (b) shows the elevation of the terrain. There are two hills in the domain: the Fourvière hill in the southwest and the Croix-Rousse hill in the northeast. The lowest elevation is 160 m. The highest elevation is 320 m.

Fig. 5 (c) represents the land surface materials in the studied domain, according to the SOLWEIG land surface classification. The ground is mostly paved or covered by grass.

Fig. 5 (d) distinguishes the various plots of the domain according to their ownership and accessibility. The private spaces, in gray, are beyond the scope for this study since the study is restricted to public areas. Within public plots, three categories are distinguished: squares and parks, roads, and closed public spaces. Squares and parks are intended to host people for long periods during hot summer days. The roads include paved areas and parking areas. They are usually used for

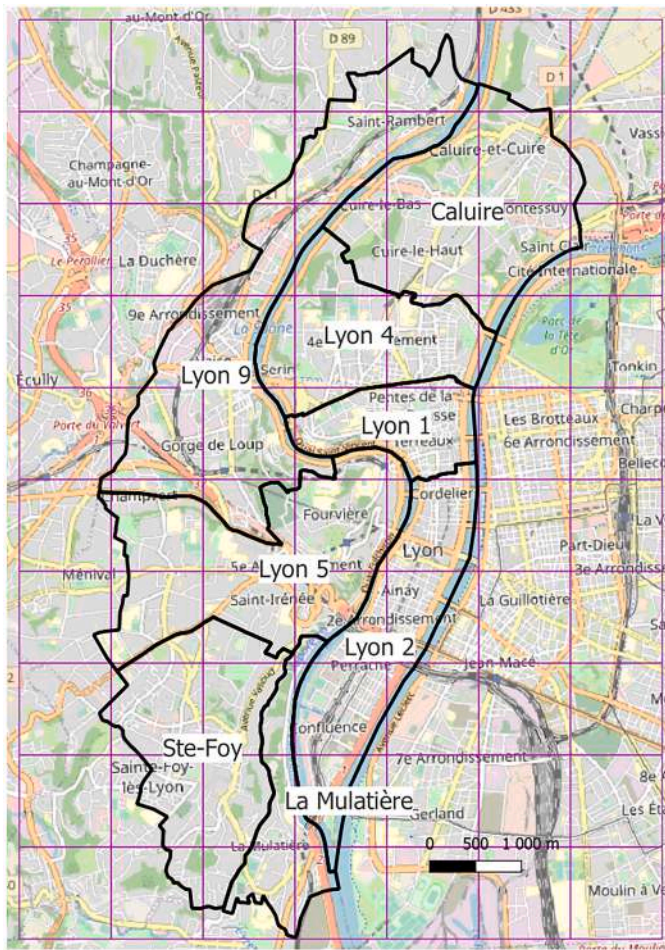


Fig. 4. Description of the study area, location within the conurbation of Lyon, limits of the municipalities, and division of the study area into square subdomains.

displacements, but they may be closed to traffic to extend the area devoted to pedestrians. Closed public spaces are either plots containing public administration buildings, including schools and gymnasiums, or the outdoor spaces that are closed to the public for safety reasons, such as the Balmes Lyonnaises. The public authorities may decide to open these areas to the public, but some construction work may be needed to ensure safe access and to prevent possible damage. We will assume that all enclosed public spaces can be made accessible to the public. The possible degradation of the thermal performance of these areas as a result of development is outside the scope of this study.

Fig. 6 shows the distribution of the land area per capita according to the category of the plots. The total area of the domain is 26 km² with 225,000 inhabitants, and each inhabitant occupies, on average, 115 m² of the land surface. Within each 115 m² unit, there are 49 m² of outdoor public space, which are made of 11 m² of squares and parks, 22 m² of roads, and 16 m² of closed public spaces. For each of these categories, the area under a tree crown is represented with a darker color. Less than 10 % of the road surface is covered by trees. In the squares and closed public spaces, approximately one-third of the land is covered by trees.

3.2. Weather data

The representative hot summer days were obtained from an analysis of the weather projection database constituted by Toesca et al. [64]. This database was collected from the CORDEX database. It contains 14 weather projections from 7 different GCM+RCM models and for 2 RCP

scenarios: RCP4.5 and RCP8.5. Each weather projection extends from 2010 to 2100. This represents a total 1260 years of weather projections for the Saint-Exupéry airport, near the Lyon conurbation.

The curves in Fig. 7 represent the July-day daily maximum temperature distributions for the three climate change stages of interest: the current state, mid-century, and end-century. Each distribution contains 8660 values: 14 weather projections, 31 days in July, 20 years periods.

Table 3 describes the daily statistics that were selected to construct the RHSD weather data. The daytime average values were computed between the average sunrise (6:06) and sunset (21:28) times in Lyon during the month of July. The daily maximum and minimum temperatures increase significantly over the century (+4.11 °C and +3.84 °C, respectively). There is no specific trend in the evolution of the average specific humidity, although the specific humidity over land is expected to increase over the century [27]. The average shortwave radiation and air to sky temperature difference slightly decrease. The average wind speed slightly increases.

The meteorological data that correspond to the three RHSDs are drawn in Fig. 8. The curves in a lighter color correspond to the current-era RHSD. The curves in a darker color correspond to the end-century RHSD. In Fig. 8, as well as in the rest of the paper, the time that is indicated under the curves corresponds to the UTC+1 time, without summer daylight saving time. Lyon is located 10.2° east from the central meridian of the GMT+1 time zone. Therefore, the sun is at its zenith at 12:50. The three global solar radiation curves are similar, with maximum radiation occurring at 13:00. The maximum air temperature values are reached at 16:00.

3.3. Simulation settings

For the SOLWEIG simulations, the albedo of the walls was set to 0.2, the average emissivity of the clothed body surface was set to $\epsilon_{cl} = 0.95$, and the absorption coefficient was set to $\alpha_{cl} = 0.7$. The latter value is an intermediate value between the absorption coefficient of dark (0.8) and light (0.6) skin colors [3], and between average absorption coefficient values (0.66–0.77) observed in Japanese students wearing summer clothes [69].

All the steps of the process were executed with a standard computer. To run the SOLWEIG simulations with a 1 m resolution, the domain was split into 1 km square subdomains, and the simulations were run separately on each subdomain. Fig. 4 shows the division of the study area into square subdomains. Five days of simulations were required to predict the mean radiant temperature over the whole studied domain during the three RHSDs.

The PET regression model was constructed for a standard male (age 23, weight 75 kg, height 1.8m), wearing light clothes (0.25clo), and resting (1met). 5th order polynomial equations were used to perform the regression. The RMSE between the reference values and the regression values was below 0.15°C for each RHSD. Fig. 9 shows a comparison between reference and regression PET values.

4. Results: global analysis

4.1. PET evolution

Fig. 10 shows the evolution of the PET in maps during the mid-century RHSD. The private spaces are beyond the scope of the study and are hidden with white patches. As expected, the PET is relatively low early in the morning and late in the evening. It reaches maximum values at approximately 15:00.

The curves in Fig. 11 show the temporal evolution of the PET distributions during the three RHSDs. The following percentiles of the spatial distributions of PET are drawn: 0 % (minimum value), 5 %, 25 %, 50 %, 75 %, 95 % and 100 % (maximum value). The air temperatures from the weather files are marked with thick black lines. The discomfort and heat stress thresholds are indicated with green lines. During

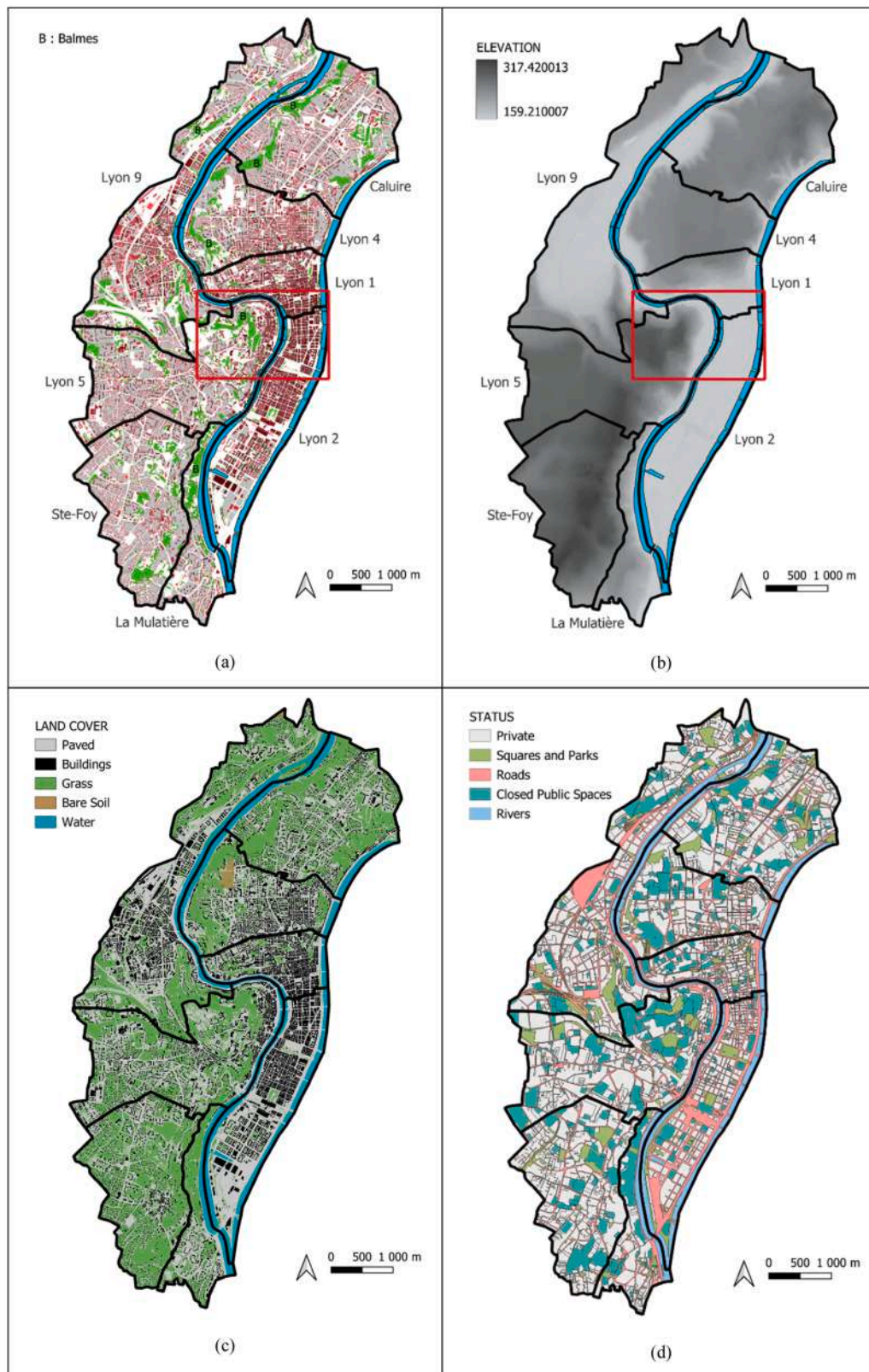


Fig. 5. Description of the study area. (a): building footprints (red) tree crowns (green disks), and private areas (grey patches in the background). (b): Elevation. (c): Land materials. (d): Land use status. The area marked in red lines corresponds to the central portion of the domain that will be used for the qualitative analysis of the results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

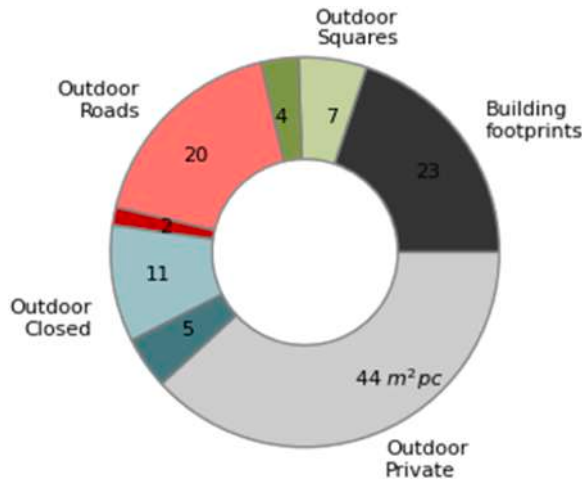


Fig. 6. Distribution of the land area per capita according to the category of the plot. All the building footprints are gathered in a single dark slice. For the outdoor public spaces (squares, roads and closed spaces), a distinction is made between the exposed areas (lighter colors) and areas under a tree crown (darker colors). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

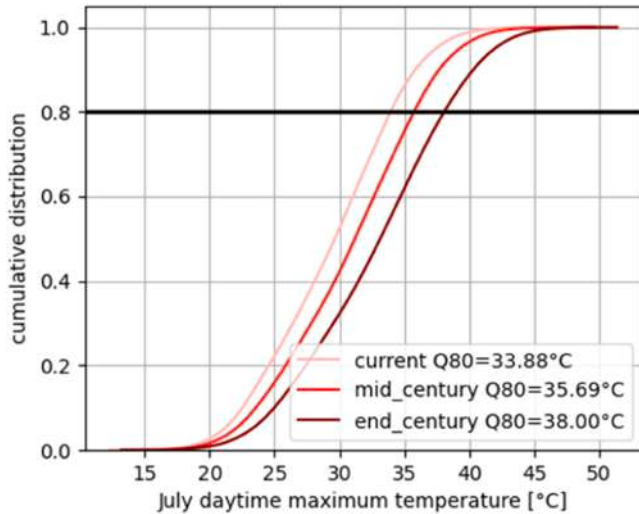


Fig. 7. Distribution of the July day maximum air temperature in the weather projection database and for the three climate change periods of interest. The horizontal line identifies the 80th percentile.

nighttime, the PET is expected to be underestimated because the urban heat island effect is not simulated. It is significantly lower than the air temperature because of the exposure to sky radiation, and because of the cooling effect of air movement. The curves indicate a considerable diversification in the outdoor thermal environments during the daytime. The amplitude of the PET spatial variations remains under 3 °C during the nighttime. It reaches 21.5 °C during the daytime.

The maps in Fig. 12 are directly obtained from the maps in Fig. 10. They show the evolution of the heat shelter status $\hat{I}_{29}$, computed with the 29 °C discomfort threshold, in the public spaces of the domain. As expected, early in the morning and late in the evening, the domain is within the fully sheltered state: all the outdoor public spaces are sheltered from thermal discomfort. The heat shelters disappear during the daytime.

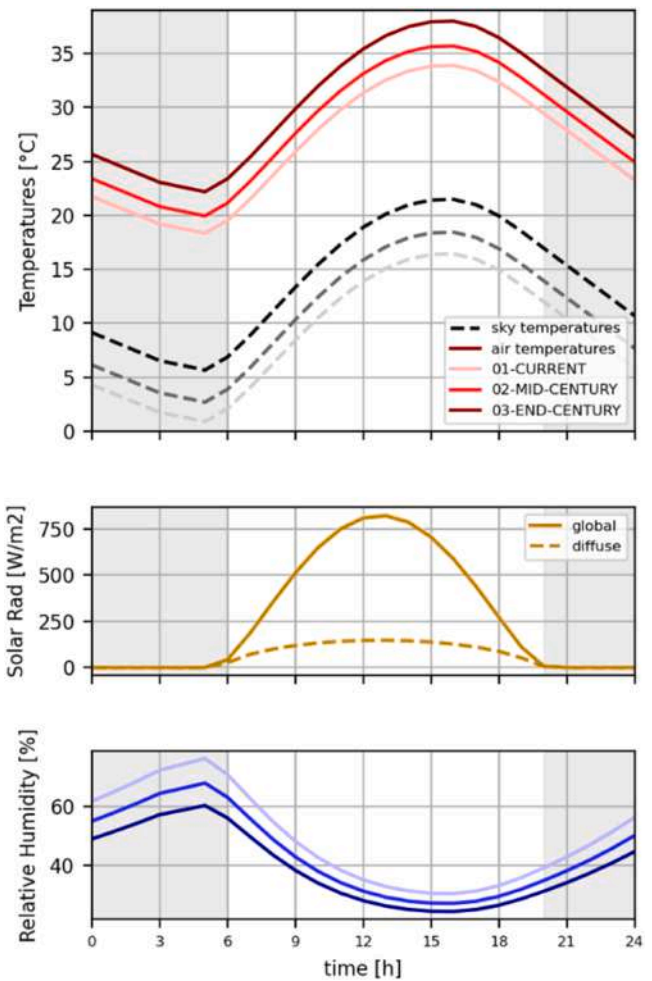


Fig. 8. Meteorological conditions during the three RHSDs. The curves in a lighter color correspond to the current-era RHSD. The curves in a darker color correspond to the end-century RHSD. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

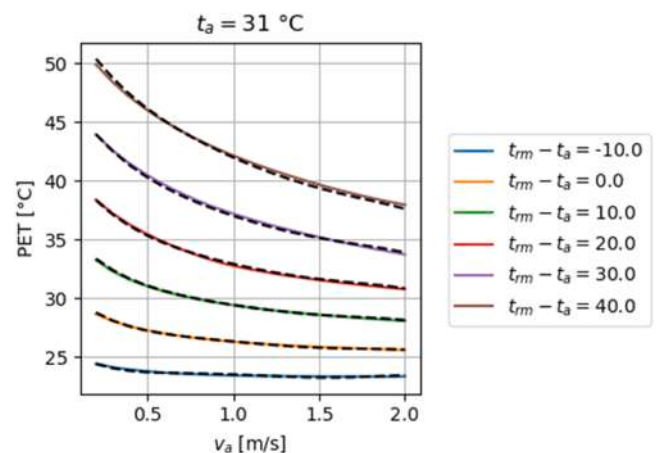


Fig. 9. Comparison between reference (solid line) and regression (black dashed lines) PET values, for the Current Period RHSD, and for $t_a = 31$ °C.

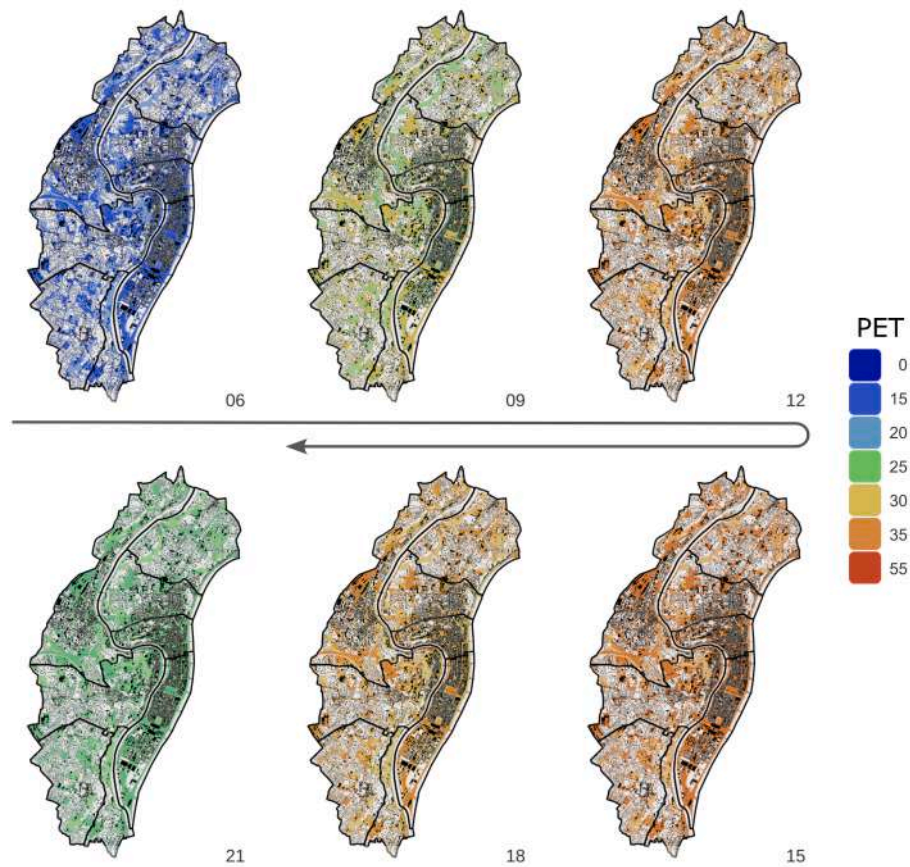


Fig. 10. Evolution of the PET distribution during the mid-century RHSD. The time of day is indicated in the lower right corner of each map. Private outdoor spaces and rivers are hidden under white patches.

4.2. Shelters from heat discomfort

Fig. 13 shows the evolution of the heat shelter area per capita $\hat{A}_{29}$ during the three RHSDs in the public spaces of the domain. The heat shelter area was computed with the discomfort threshold (29 °C): the heat shelters protect the occupants from discomfort. A distinction is made between the squares and parks (green), the roads (red), and the

closed public spaces (peacock blue). The areas that are filled with colors in the graphs represent the areas that are used to compute the heat sheltering performance; they are restricted to the daytime period.

The shape of the curves is similar to the shape of the curves expected in Section 2.5. During the nighttime periods, the domain is fully sheltered, and the heat shelter area per capita is equal to the area of the domain: 49 m²pc. During the daytime, the heat shelter area per capita is reduced.

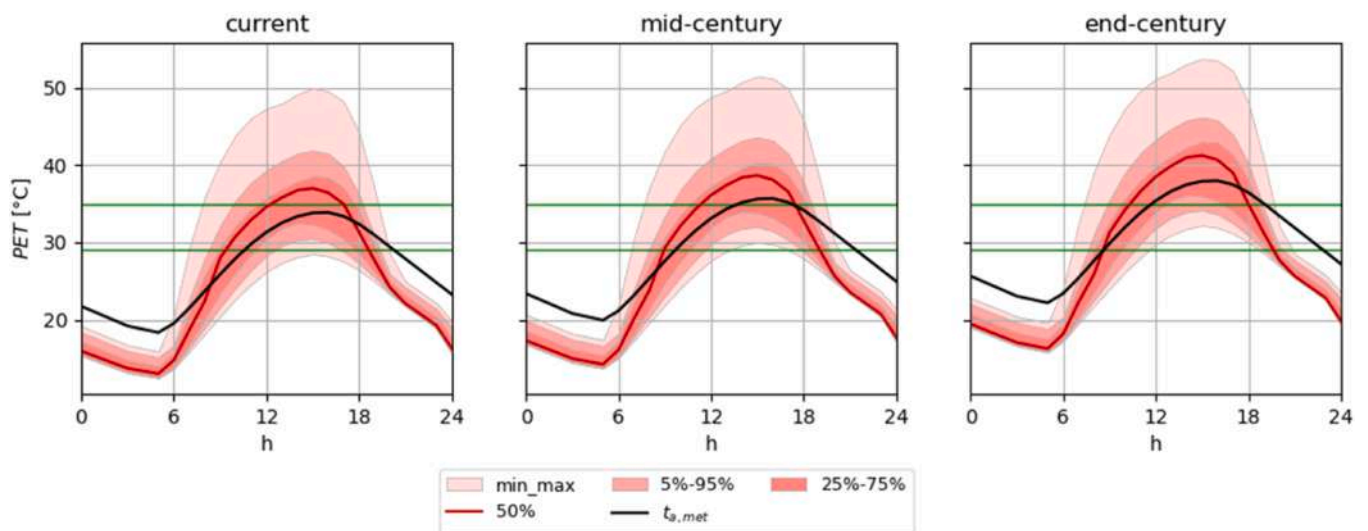


Fig. 11. Evolution of the PET distribution in the public outdoor spaces of the study area during the three RHSDs. The air temperatures given at the meteorological station are drawn with black curves. The green horizontal lines represent the discomfort and the heat stress thresholds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

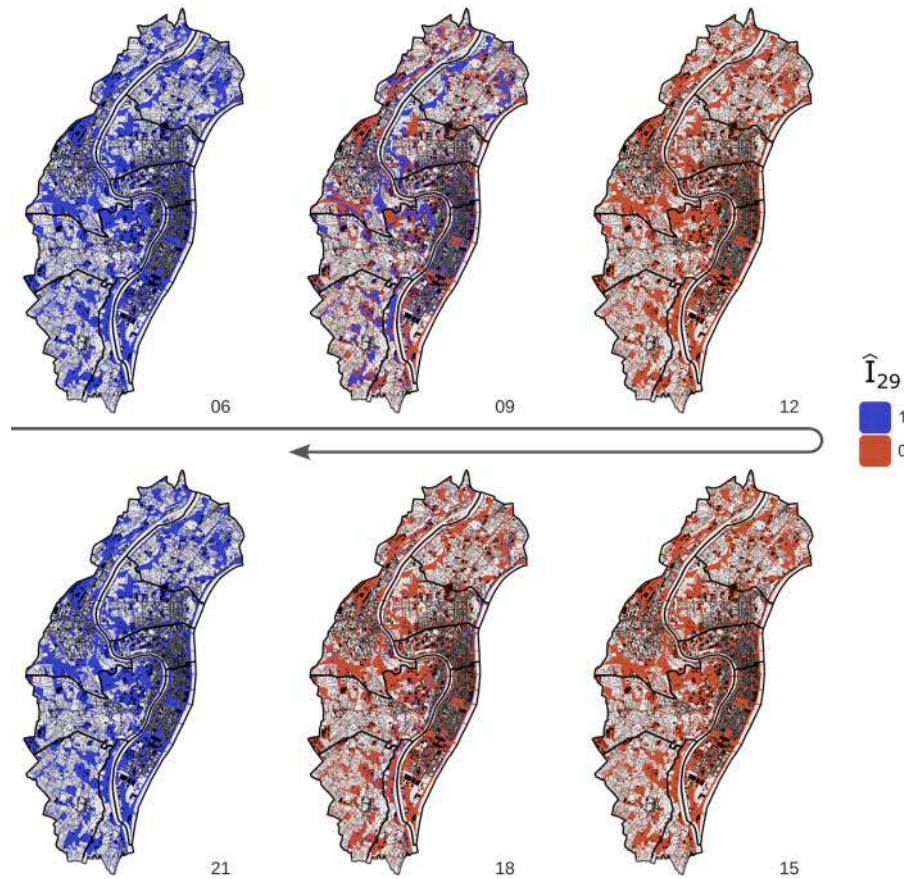


Fig. 12. Evolution of the distribution of heat shelters at the discomfort threshold during the mid-century RHSD.

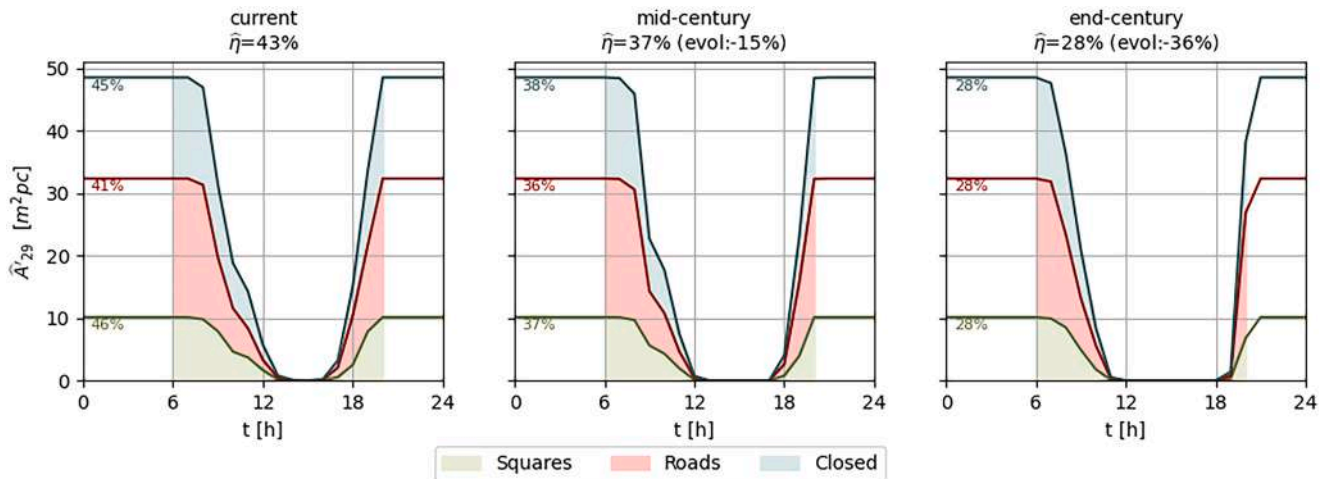


Fig. 13. Evolution of the heat shelter area per capita $\hat{A}_{29}$ for protection against heat discomfort in the public outdoor spaces of the whole domain during the three RHSDs.

The curves reveal that during the three RHSDs, there is a period during which the area covered by the heat shelters is zero. The domain is thus in the “fully exposed” state. There is no outdoor shelter from heat discomfort in any part of the domain. The fully exposed state lasts for 2 h during the current era RHSD, 4 h during the mid-century RHSD, and 6 hours during the end-century RHSD.

Such long full exposure events help us understand the magnitude of the issues related to outdoor thermal environments during hot summer days. During hot summer afternoons, searching for thermal comfort in outdoor spaces is hopeless. The urban fabric is not able to provide

comfortable places. There is no public outdoor heat shelter that can protect occupants against heat discomfort.

4.3. Shelters from heat stress

Fig. 14 shows the evolution of the heat shelter area per capita $\hat{A}'_{35}$ during the three RHSDs in the public spaces of the domain. Here, the heat shelter area was computed with the heat stress threshold (35°C). The heat sheltering performance values $\hat{\eta}_{35}$ for all the outdoor public

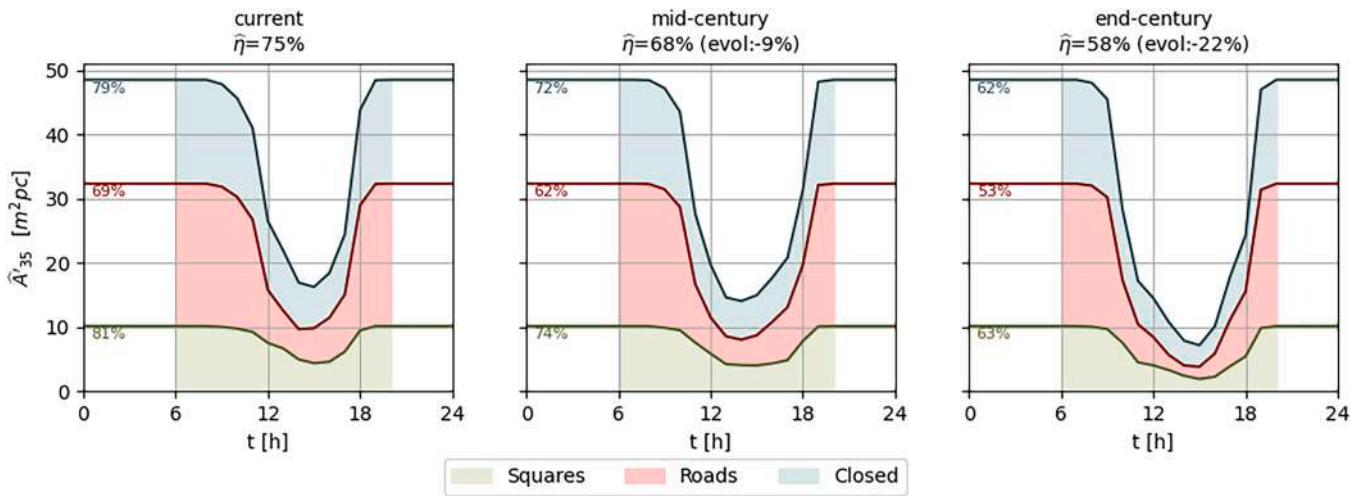


Fig. 14. Evolution of the heat stress shelter area per capita $\hat{A}_{35}$ in the public outdoor spaces of the whole domain during the three RHSDs. The percentage at the beginning of each curve represents the heat sheltering performance for the related type of outdoor place.

spaces are given at the top of the subplots. The heat sheltering performance for each category of outdoor public space is indicated at the beginning of each curve.

The heat sheltering performance $\hat{\eta}_{35}$ is equal to 75 % during the current era RHSD. This means that, on average, through the daytime period, 75 % of the outdoor public area, including closed spaces, can shelter the population from heat stress. This performance is reduced by 9 % during the mid-century RHSD and 22 % during the end-century RHSD. The roads are systematically 10 % less effective at providing heat shelters than the other categories of outdoor public spaces. For example, during the end-century RHSD, the heat sheltering performance is equal to 53 % on the roads. It is equal to 62–63 % in the squares and in the closed public spaces.

The curves pass through a minimum value of $\hat{A}_{35}$ during the afternoon. This is a critical point, because it corresponds to a state of the urban tissue for which the availability of heat shelters in the outdoor areas is minimum. The coordinates of the critical point (critical time, and critical heat shelter area) are given in Table 4 for the three RHSDs. The critical time is 14:00 or 15:00. It lies between the time with maximum solar radiation (13:00) and the time with maximum air temperature (16:00). The critical heat shelter area decreases during the century. It goes from 35 % of the total outdoor area during the current era RHSD, to 15 % of the total outdoor area at the end of the century. At the end of the century, there will only be 7.2 m² of heat shelter available per citizen during the critical time.

There is currently no absolute threshold from which the heat shelter area per capita $\hat{A}_{35}$ may be considered sufficient. This issue will be discussed later in the paper, but let us consider 10 m²pc as a reasonable arbitrary threshold. Table 2 shows the time intervals within the three RHSDs during which the heat shelter area per capita $\hat{A}_{35}$ drops below this threshold. Two cases are distinguished: in the first case, the closed public spaces (peacock blue areas in Fig. 14) remain inaccessible, and in the second case, access is arranged to all the closed public spaces.

For the current era RHSD and the mid-century RHSD, arranging access to closed public spaces would allow us to reach 10 m²pc during the whole duration of the hot summer day. However, if the closed public places remain inaccessible, there is systematically a period of the day during which the heat shelter area drops below 10 m²pc. During the end-century RHSD, the heat shelter area consistently drops below 10 m²pc during the daytime. Arranging access to the closed public space would reduce the duration of the time interval during which the heat shelter area is below the threshold. It would also provide extra areas where the occupants may rest during the morning and during the evening.

Table 2
Mapped data sources.

Quantity	Source Name	URL
Buildings	BDTOPO	www.data.gouv.fr/fr/datasets/sig-bdt-opo-2/
Elevation	Modèle Numérique de Terrain	www.data.gouv.fr/fr/datasets/mnt-2014-raster/
Street Trees	Data Grand Lyon	https://data.grandlyon.com/jeux-de-donnees/arbres-alignement-metropole-lyon/info
Land Cover	Corine Land Cover	https://data.grandlyon.com/jeux-de-donnees/classification-espaces-vegetalisees-artificialisees-production-corine-land-cover-metropole-lyon-ortho-2015
Land Cover	Data Grand Lyon/[4]	https://data.grandlyon.com/jeux-de-donnees/vegetation-stratifiee-2018-metropole-lyon
Administrative Status	Cadastre	https://www.geoportail.gouv.fr/donnees/parcelles-cadastrales
Population	INSEE	https://www.insee.fr/fr/statistiques/4799309

5. Results: qualitative analysis

The studied domain is sufficiently large to exhibit a great diversity of urban layouts. The qualitative exploration of heat shelter maps aims to identify which of these urban layouts are the most efficient for producing heat shelters.

In this section, qualitative analysis is performed over a central portion of the domain during the end-century RHSD. The location of the central portion of the domain is indicated with a red rectangle in Fig. 5 (a). Fig. 15 shows the elevation, building footprints, and tree crown

Table 3
Selected daily statistics used to reconstruct the RHSD meteorological data.

	Current	Mid-century	End-century
Maximum temperature [°C]	33.88	35.69	38.00
Minimum temperature [°C]	18.33	19.93	22.17
Average wind speed [m/s]	2.46	2.56	2.65
Average specific humidity [g/kg]	9.94	9.78	9.97
Average shortwave radiation [W/m ²]	470.2	467.3	465.9
Average air to sky temperature difference [°C]	17.45	17.26	16.52

Table 4

Coordinates of the critical point. Time intervals during which the heat shelter area per capita $\hat{A}_{35}$ is below $10 \text{ m}^2\text{pc}$.

		RHSD		
		Current period	Mid-Century	End-century
Critical point	Hour	15:00	14:00	15:00
	$\hat{A}_{35} [\text{m}^2\text{pc}]$ (%) total area)	16.9 (34 %)	14.1 (29 %)	7.2 (15 %)
$\hat{A}_{35} < 10 \text{ m}^2\text{pc}$	No access to closed spaces	14:00-15:00	12:30-15:30	11:15-16:45
	Full access to closed spaces	x	x	13:15-16:00

occupation in this area. The central portion of the domain includes, from the right to the left, the densely built central district of Lyon, the S-shaped Saône River, the densely built medieval district, the Fourvière hill side with the Balmes Lyonnaises (large patches of continuous tree cover), and the Saint-Just district with many private outdoor spaces and less dense urbanization.

Fig. 16 shows the evolution of the heat shelter status maps over the central portion of the domain during the end-century RHSD. The heat shelter status $\hat{I}_{35}$ was computed with the heat stress threshold. The heat shelters are identified with blue patches. The arrows in the lower right corner of the pictures indicate the direction of the sun. The maps in Fig. 16 correspond to the evolution of the heat shelter area per capita drawn in the right-side subplot of Fig. 14.

Before 9:00, the entire domain is sheltered from heat stress. At 9:00, the areas exposed to heat stress are located in the central parts of east/west oriented streets (solid line rectangle), which combine low wind speed and exposure to direct sun radiation at pedestrian level. Exposed areas are also located at the foot of sunlit walls without overlooks (dashed line rectangles), where the exposure to sun radiation is



Fig. 15. Central portion of the studied domain, over which the results of the qualitative analysis are shown. Top map: building footprints in red, with a darker color for the higher buildings, tree crowns in green, with a darker color for higher trees. The grey patches represent private spaces. Bottom map: Elevation in grayscale. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

intensified by the reflection of radiation over the whole surface of the walls.

Between 9:00 and 11:00, there is a sharp decrease in $\hat{A}_{35}$ (Fig. 14): air temperature and solar radiation reach values such that the PET exceeds the heat stress threshold in all areas exposed to direct solar radiation, despite high wind speeds. At 11:00, the domain is in the “Shelter = Shade” state: all the areas that are shaded from direct sun radiation seem to be sheltered from heat stress, and all the areas that are exposed to direct sun radiation seem to be exposed to heat stress. The outdoor spaces are either shaded by trees or by buildings.

After 11:00, the decrease in $\hat{A}_{35}$ slows down. This decrease is now due both to the sun’s displacement, which reduces the shaded area, and to the increase in air temperature and solar radiation, which makes shaded areas more vulnerable to heat stress. The areas that are both sufficiently confined for wind speed to be low, and sufficiently open to sky for pedestrians to be exposed to solar radiation are particularly vulnerable to heat stress. This is why, between 14:00 and 16:00, almost all canyon-shaped streets are exposed to heat stress.

Between 14:00 and 16:00, the vast majority of heat shelters are located under large patches of continuous tree cover (dashed-dots line polygons). But within these patches of continuous tree cover, the area protected from heat stress gradually shrinks over the course of the afternoon. A surrounding buffer zone exposed to indirect solar radiation progressively falls into the state of heat stress. In small patches of tree cover (solid line polygons), this buffer zone does not leave much space for heat shelters.

Between 14:00 and 16:00, the second major source of heat shelters is located at the foot of shaded walls with little overlook (dashed line polygons). These locations are sufficiently open to be exposed to wind. They are sufficiently far from sunlit façades that would expose them to reflected sun radiation. Since the sun direction evolves during the day, the location of these heat shelters evolves. At 16:00, the only canyon-shaped street in which heat shelters reappear periodically is the only street that hosts a significant population of trees (solid line rectangle).

At 18:00, the “shelter = shade” state is recovered. Once again, there is a direct correspondence between the sheltered areas and the shaded areas. After 18:00, the area covered by heat shelters increases due to the sun displacement and the decrease in the air temperature. The fully sheltered state is recovered at 20:00 (Fig. 14).

To summarize, the sheltering state of the domain goes through a cycle with a “dual thermal constraint” period surrounded by two “shelter=shade” states. The “dual thermal” constraint period is marked by the restricted distribution of shaded areas and their reduced capacity to buffer against thermal extremes, which reduce the sheltered area to the core of large patches of continuous tree cover and to the foot of shaded façades without overlooks.

Many studies have already demonstrated the capacity of trees to offer moderate spaces in term of heat loads [16,24,38]. But, concerning the performances of the street layouts, it is difficult to relate the present results to the existing knowledge. A comprehensive review from Jamei and Rajagopalan [30] reports general agreements that the east/west streets present the worst conditions, as they suffer from prolonged exposure to solar radiation, and that deep and narrow streets are more efficient as they are less exposed to the sun. Here, during the core of the day, no clear relationship can be established between the orientation of the streets, their narrowness and their capacity to produce heat shelters: there is almost no heat shelter in the canyon-shaped streets. This apparent discrepancy between the present results and the existing knowledge may be due to the particular strength of the weather conditions employed for the simulations, and to the non-linear nature of the heat shelter identification method: even if some streets perform better than other streets, they all fall into the heat exposed category. Heat shelters are only found where there is no opposite façade to block the wind and reflect radiation.

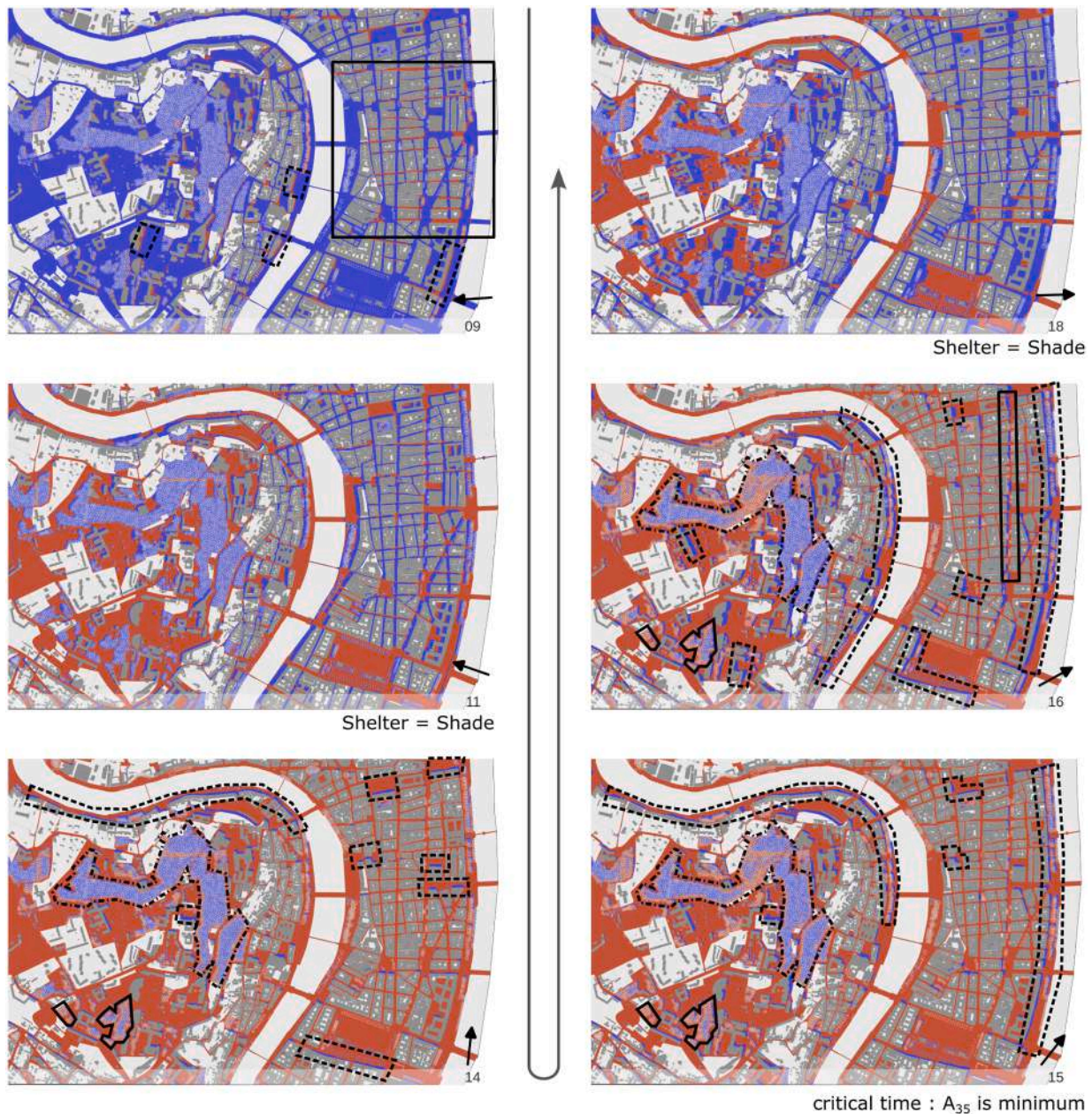


Fig. 16. Location of the heat shelters (heat-stress threshold) during the end-century RHSD. The heat shelters are indicated with the blue color. The private properties are hidden using white polygons. The trees are indicated with transparent white disks. The areas of particular interest are highlighted with black surroundings. The time of day and the direction of the sun are indicated in the lower right corner of each map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

7. Results: disparities between municipalities

The evaluation of the disparities between the municipalities of the domain is based on the results that are drawn in Fig. 17. The figure contains curves that represent the evolution of heat shelter area per capita $\hat{A}_{35}$ in the public areas of each municipality during the current-era RHSD and the end-century RHSD. As in Figs. 13 and 14, the squares, the roads, and the closed outdoor public spaces are distinguished. The scale is the same for each curve to allow for a direct comparison between the municipalities and the RHSDs. On the top of each plot, two values of the daytime average heat shelter area per capita $\hat{A}_{DT}$ are given: one for the accessible public spaces (roads + squares and parks), and one for the closed public spaces. The heat sheltering performance $\hat{\eta}_{35}$ is given separately for each category of outdoor public

spaces (values at the beginning of each curve).

Fig. 17 also contains pie charts that show, for each municipality, the distribution of the land surface according to the land-use category: building footprints (black color), private outdoor areas (gray color), and public outdoor areas, which are distinguished according to their type (in green, red, and peacock blue) and their tree cover state (darkness of the color). These pie charts are similar to the pie chart drawn in Fig. 6. They reveal two groups of municipalities whose configurations are diametrically opposed. On the one hand, the municipalities of Lyon 1 and Lyon 4 host a dense population of buildings. Their public outdoor spaces are mostly roads, and most of the trees are located on these roads. On the other hand, Sainte-Foy, la Mulatière and Caluire have a lower building density. There are almost no trees in their streets. In these municipalities, there are plenty of closed public spaces that are heavily planted.

The disparities between the municipalities, in terms of heat shelters

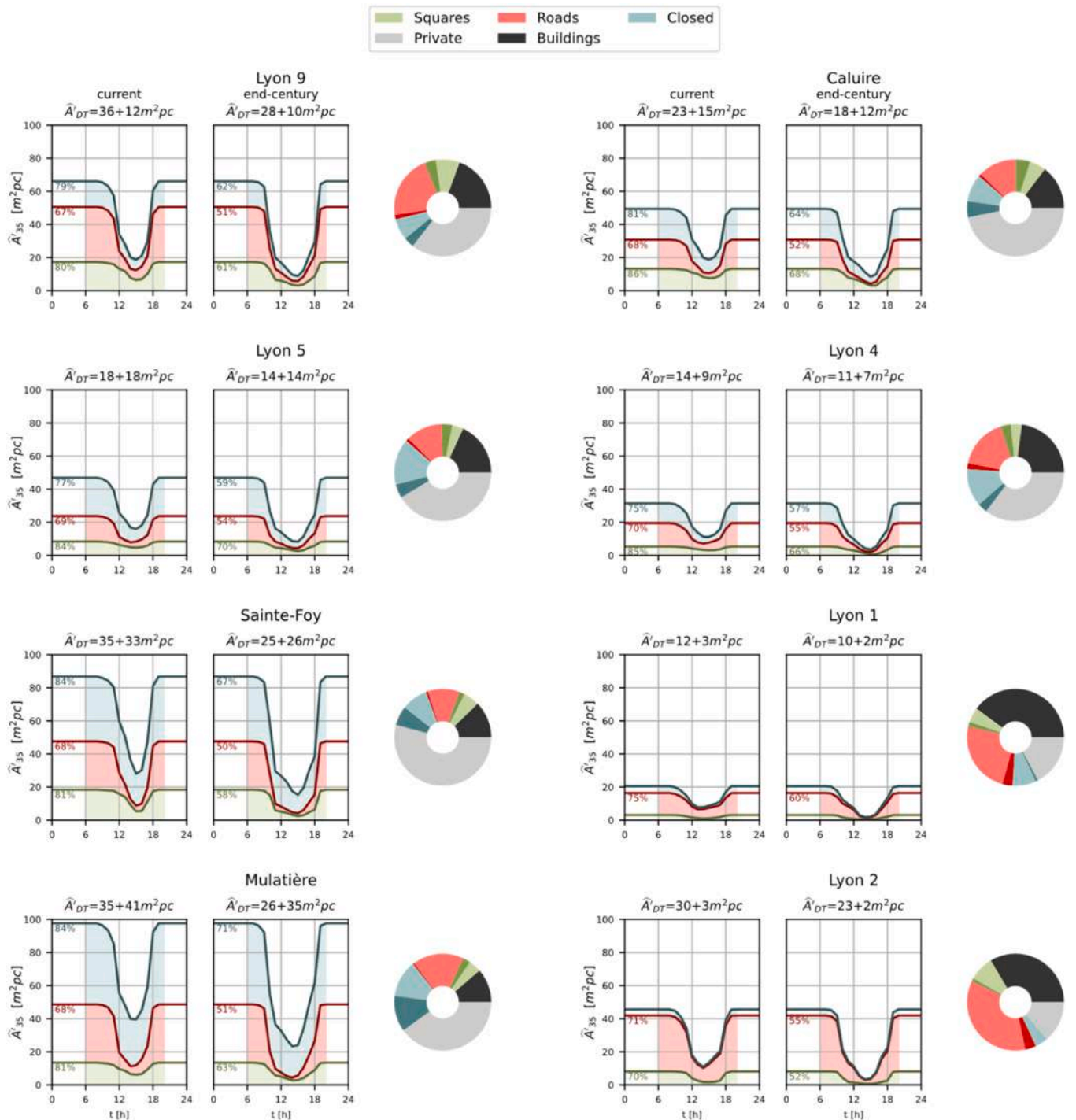


Fig. 17. Heat stress shelter area per capita $\hat{A}_{35}$ in the various municipalities of the domain. The pie charts represent the distribution of the ground area according to the land status for each municipality. On top of each plot: daily average heat shelter area for accessible and closed public spaces. At the beginning of each curve, heat sheltering performance for the corresponding space category.

availability, are first assessed through the values of the daytime average heat shelter areas $\hat{A}_{DT}$ in the outdoor accessible public spaces (first values in the title of each curve). During the current era RHSD, the surface of the accessible heat shelters is more than twice higher in Lyon 9, Sainte-Foy, and la Mulatière (30 to 35 m^2pc) than in Lyon 1 and Lyon 4 (12 to 14 m^2pc). This ratio is kept during the end-century RHSD (23 to 26 m^2pc as opposed to 10 to 11 m^2pc). These disparities are primarily

explained by inequalities in terms of outdoor available public spaces. In Lyon 1 and Lyon 4, there is only 18 to 20 m^2pc of outdoor available public spaces that are accessible to the population. There is nearly 50 m^2pc of outdoor available public spaces in Lyon 9, Sainte-Foy and La Mulatière.

The reservoir of closed outdoor public spaces (peacock blue curves) is also unevenly distributed: it is nearly null in Lyon 1 and Lyon 4 and reaches 49 m^2pc in La Mulatière and Sainte-Foy. In the latter munici-

palities, the closed outdoor public spaces may bring a considerable amount of heat shelters during the current era RHSD (33 to 41 m²pc), and during the end-century RHSD (26 to 35 m²pc). If access had to be arranged in these closed spaces, it may increase the disparities in terms of heat shelter availability over the territory.

The critical heat shelter area, i.e., the minimum value of the heat shelter area during the day, differs among the municipalities. At the end of the century, if an access is arranged to the closed public spaces, the critical heat shelter area is about 3 m²pc to 5 m²pc in Lyon 1, Lyon 2 and Lyon 4. It is about 10 m²pc in Caluire, Lyon 9, and Lyon 5. It will reach values between 17 m²pc and 22 m²pc in La Mulatière. In some municipalities, there will be 8 times more outdoor heat shelters as in others at the critical time.

In the squares, parks, and closed public spaces, the heat sheltering performance $\hat{\eta}_{35}$ is generally, but not systematically, higher than that in the streets. It ranges between 70 % and 86 % during the current-era RHSD and between 52 % and 71 % during the end-century RHSD. In these areas, most of the heat shelters are due to the shading effect of trees. Fig. 18 shows the relationship between the tree-cover fraction λ_{TC} and the heat sheltering performance $\hat{\eta}_{35}$ during the three RHSDs. The tree-cover fraction is the proportion of the outdoor land surface that is located under a tree crown. The curves show that once λ_{TC} has reached 20 %, the relationship between $\hat{\eta}_{35}$ and λ_{TC} is nearly linear. For example, during the current-era RHSD, the relationship between $\hat{\eta}_{35}$ and λ_{TC} can be approximated with the following equation:

$$\hat{\eta}_{35} \approx 0.32 \cdot \lambda_{TC} + 69.3 \quad (7)$$

Thus, during the hot summer days of the current era, an increase in the tree-cover fraction of 1% corresponds to an increase in the heat shelter performance of 0.32 %. In other words, adding 1 m² to the area of the tree crown would result in 0.32 m² additional heat shelter area on average during the daytime. This ratio increases to 0.41 during the mid-century RHSD and to 0.43 during the end-century RHSD.

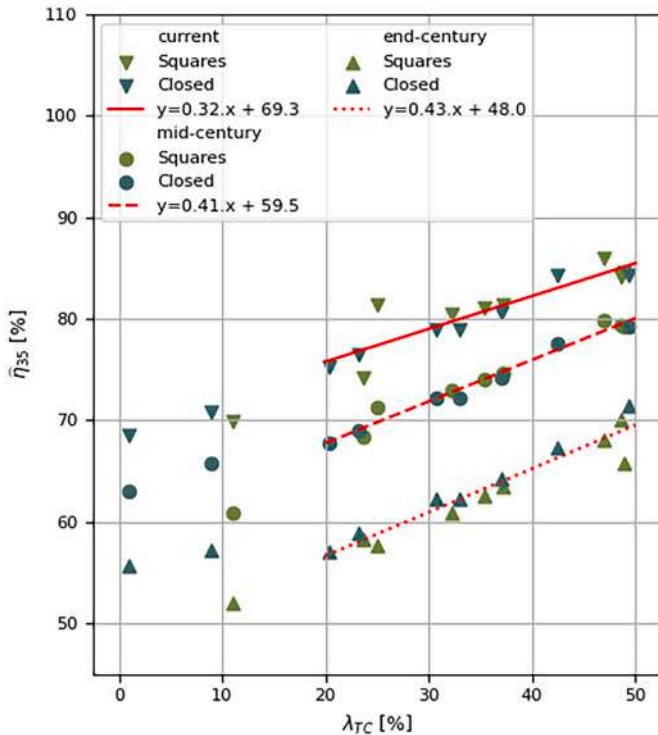


Fig. 18. Relationship between the heat sheltering performance $\hat{\eta}_{35}$ and the tree-cover fraction λ_{TC} in squares and parks, and closed public spaces. Each point corresponds to one municipality.

8. Limitations of the PET calculation procedure

The PET calculation is based on a number of assumptions that have enabled the calculation to be carried out over a wide study area. This section is about the uncertainties on PET that these assumptions may cause.

The local wind speed was computed with the regression curve shown in Fig. 2. This regression curve represents an average trend in wind speed evolution: the more open the area, the more exposed it is to wind speed. However, there is a considerable dispersion of reference points around the average trend. This dispersion may be the result of local channeling and blocking effects, due to complex three-dimensional flows that depend on the wind direction and the specific shape of the urban blocks.

The dispersion of the wind velocity factor WVF around the regression curve has been quantified in the form of a 95 % confidence interval, assuming that the error follows a normal distribution. The width of the confidence interval has been corrected for low SVF values to avoid negative WVF . The resulting 95 % confidence interval is drawn in Fig. 2. A Monte-Carlo technique has then been applied to propagate the WVF dispersion into the PET computation functions. Preliminary tests showed that the PET confidence interval $\Delta^{95}PET$ was highly sensitive to SVF and t_a . Fig. 19 shows the dependency between $\Delta^{95}PET$ and these two parameters.

The maximum values of $\Delta^{95}PET$ is around 5 °C. It is reached when $t_a < 30$ °C, and when $SVF \approx 0.5$, which corresponds to intermediate sky exposures. This uncertainty is not negligible. It may constitute a major limitation of the PET computation process proposed by the present study. Therefore, a more advanced wind speed prediction method would be required to improve the methodology. The recent URock model [5] may be a good candidate for replacing the regression technique. Its usage over large domains still remains to be tested.

In addition, the protocol assumes that the air temperature is homogeneous over the urban tissue. In cities, air temperature variations may be due to the UHI effect, to adiabatic expansion of air, or to microclimatic effects. As mentioned in the introduction, field measurement studies suggest that the effect of UHI is negligible during the day in non-coastal cities, but there is no theory assess this behavior is systematic. Therefore, some knowledge about the UHI pattern in the city of interest would be required to confirm this assumption before using the protocol. In Lyon, measurements from Toesca et al. [65] confirmed that UHI had no effect on daytime air temperatures. Assuming a vertical gradient of 0.7 °C/100 m, adiabatic expansion of air would have caused roughly a 1

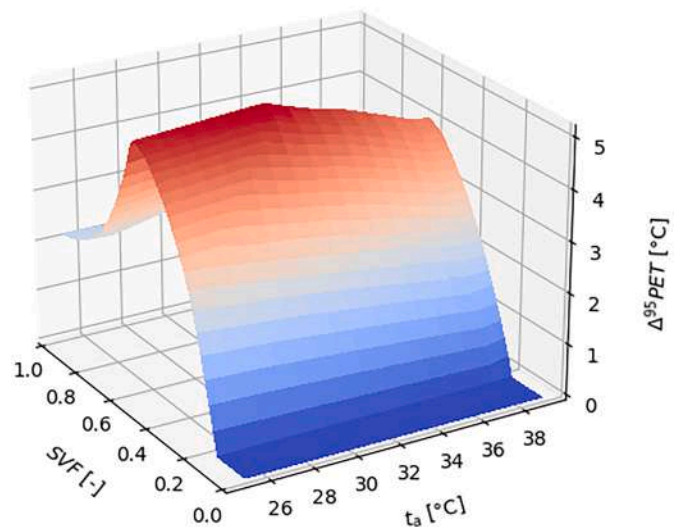


Fig. 19. PET confidence interval as a function of the air temperature and SVF , for $RH = 50\%$, $t_{rm} - t_a = 30$ °C, and $v_{a,met} = 2.5$ m/s.

°C air temperature variation over the domain.

Air temperature variations due to local microclimatic effects have been reported in the literature. For example, Ma et al. [43] predicted air temperature reductions up to 2.3 °C in some narrow street canyons of Hong Kong, Skoulika et al. [63] observed a 3.3 °C air temperature reduction in parks of Athens, Ng et al. [56] reported a 0.5 °C air temperature reduction due to street trees. These effects may affect the results of the study, especially the results obtained in narrow streets. But capturing these effects would require using models that are incompatible with the size of the studied domain. Therefore, new efficient simulation models for predicting these effects would be required to better handle air temperature variations.

9. Conclusions and outlook

The present paper proposes a procedure for evaluating, both qualitatively and quantitatively, the availability of heat shelters in urban areas. The procedure does not intend to introduce any breakthrough model or concept. It is a coherent assembly of an input weather processing method, a simulation protocol that permits to compute the PET over the studied domain, and some indicators that are intended to provide answers to the five questions listed in the introduction. The input weather processing method and the simulation procedure are coherent with the scale of the domain that has to be investigated. The definition of the output indicators and the analysis workflow are coherent with the nature of the information that is needed to answer the five questions listed in the introduction.

The procedure was applied to evaluate the availability of the heat shelters in the Lyon conurbation. Below is a nonexhaustive list of the major conclusions from this evaluation. Each conclusion provides a partial answer to one or more of the five questions listed in the introduction. The number of each associated question is given at the end of each item:

- Comfortable conditions are not expected in outdoor public heat shelters during hot summer days. Heat shelters will, at most, protect the population from heat stress. (Q2)
- The heat shelter area will drop below 10 m²pc repeatedly. At the end of the century, during the most critical time of the day, the urban tissue will only be able to provide 7.2 m² of heat shelter per inhabitant in the outdoor public spaces, and this outdoor heat shelter area will be unevenly distributed among the various municipalities, varying between 3 m²pc in densely build municipalities and 20 m²pc in municipalities with less buildings and more trees. (Q1,3,5)
- The thermal state of the domain goes through a cycle with a “dual thermal constraint” period surrounded by two “shelter=shade” states. During the “dual thermal constraint” period, the heat shelters are exclusively located within shaded areas with sufficient ventilation and low exposure to reflected solar radiation. These areas are located under large patches of continuous tree cover, and at the foot of shaded building façades without overlooks. (Q4)
- In the plots that host lots of trees, the daytime average heat shelter area evolves linearly with the tree crown area. During hot summer days of the current period, each 1 m² of a tree crown provides on average 0.32 m² of heat shelter area. (Q3,4)

The application of the heat shelter evaluation protocol to the conurbation of Lyon has demonstrated the ability of the protocol to provide answers to the five questions listed in the introduction. Additionally, from the authors' point of view, the scope of some conclusions goes beyond the Lyon conurbation case study. Indeed, in many cities in the world, the climate and the urban morphology are similar to the climate and the urban morphology in Lyon. The conclusions of the evaluation may be extrapolated to these cities.

Furthermore, the heat shelter evaluation raised some issues that

should be investigated further by the scientific community. For example, the “shelter = shade” states were identified with a qualitative observation of the maps. A more rigorous approach, with appropriate mathematical tools, should be developed to identify the shelter=shade state. Additionally, it would be interesting to assess the occurrence of the “shelter = shade” state in various cities under various climates. Similarly, the occurrence of a linear relationship between the tree crown area and the average heat shelter area in open public spaces should be challenged in various cities and various climates.

The qualitative results invite to rethink the way urban layouts should be designed for heat mitigation. The urban layouts must provide shade, the shaded areas should not be exposed to the solar radiation reflected by the surrounding surfaces, and they should be well ventilated. Large patches of trees seem to fulfill these criteria. Building shade only provides heat shelters if the façade has no overlooks. The studied domain did not offer the possibility to investigate the performance of the combined effects of trees and building shades.

In the present study, the predicted evolution of the heat shelters availability across the century was only due to variations of the meteorological loads. The evaluation was made without considering the dynamic development of the city across the century. If such a development has to be taken into account, tree growth models and urban development projections will have to be included into the methodology.

Finally, the heat shelter evaluation protocol uses the heat shelter area per capita to quantify the availability of the heat shelters. This indicator may be the simplest indicator for that purpose. It is easy to imagine what the value of this indicator represents. However, the indicator faces a major issue: there is currently no threshold that would indicate when the quantity of heat shelters is sufficient. To define such a threshold, further research may be conducted in various research fields.

First, the heat shelter area has been calculated without taking into consideration that a part of the land surface is already occupied by urban furniture, cars, tree trunks, and bushes, etc. A technique to extract the space available for citizens to rest on a large domain must be developed. The population count was solely based on the number of inhabitants. This count could be refined by considering daily mobilities. Datasets are already available for this purpose in several cities around the world [66]. Then, social studies should be conducted to determine how much space is needed for occupants to feel comfortable and safe in outdoor areas. Further social studies must also be conducted to determine the number of people who may stay simultaneously in heat shelters. To determine such a number, researchers will have to identify the reasons that may motivate citizens to take refuge in outdoor public heat shelters. Indeed, these shelters protect against heat stress, but they are uncomfortable. People would stay in such uncomfortable spaces only if they don't have any other choice. In the middle of the day, buildings may be better options than outdoor public spaces for finding bearable thermal environments. Such results suggest the necessity of adapting the way people live during the summer in future decades.

CRediT authorship contribution statement

Damien David: Writing – review & editing, Writing – original draft, Visualization, Supervision, Data curation, Conceptualization. **Marjorie Salles:** Visualization, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Damien DAVID reports financial support was provided by Centre for Energy and Thermal Sciences of Lyon. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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